

TAMIL NADU STATE BOARD - STANDARD 9 MATHEMATICS

CHAPTER 8: STATISTICS

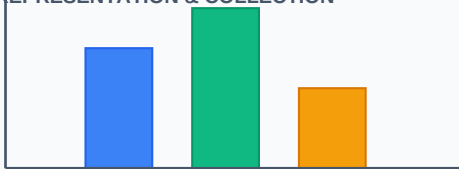
PREMIUM ACADEMIC STUDY SUITE & EXAM GUIDE

Board:	Tamil Nadu State Board (TNSB)
Syllabus:	Samacheer Kalvi Textbook Curriculum aligned
Standard:	Class 9 English Medium
Subject:	Mathematics
Chapter Name:	Statistics

Learning Objectives:

- Comprehend core theoretical concepts and exact textbook terminology.
- Apply relevant formulas and proofs to solve high-scoring board problems.
- Interpret and sketch vector diagrams or graphical models representing equations.
- Build examination readiness for 1-mark, 2-mark, and 5-mark question sets.

UNIT ROADMAP: DATA REPRESENTATION & COLLECTION



Core Concepts & Definitions

Textbook definitions are essential for scoring fully in the short-answer section of board assessments. Review the primary definitions below:

Term	Official Definition & Context
Primary Data	Data collected directly from first-hand sources by the investigator for a specific research purpose.
Arithmetic Mean	The average value of a set of numbers, calculated by dividing the sum of all terms by their count.
Median	The middle value in an ordered set of data, separating the higher half from the lower half.
Mode	The value that appears most frequently in a given dataset. A dataset can have one mode, multiple, or none.

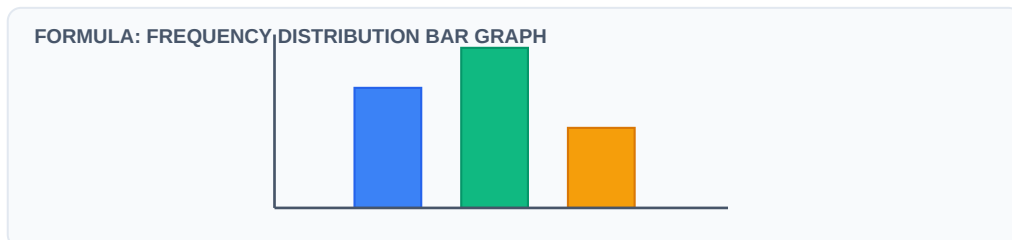
Syllabus Exam Tip: Memorize precise keywords. Examiners award maximum marks when textbook terms and correct symbols are correctly referenced.

GLOSSARY: DATA ORGANIZATION STAGES



Detailed Explanation & Formulas

Here, we analyze the core laws and formulas. Examine the conceptual illustration below to visual the relation between variables:



Essential Formulas, Laws & Identities:

- Arithmetic Mean (Ungrouped): $\text{Mean} = \frac{\sum x}{n}$
- Mean (Grouped Direct): $\text{Mean} = \frac{\sum fx}{\sum f}$
- Median (n is odd): term at index $(n + 1)/2$
- Median (n is even): average of terms at index $n/2$ and $(n/2) + 1$

Make sure to check the dimensional consistency of all physical quantities and verify that units are correctly stated in final solutions.

Worked Examples

Review these standard board examination-style worked examples to learn proper step layout structures:

Example 1: Find the mean of the numbers: 2, 4, 6, 8, 10.

Step-by-step Solution:

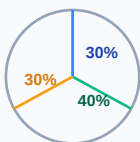
Sum of numbers = $2 + 4 + 6 + 8 + 10 = 30$. Count $n = 5$. Mean = $30 / 5 = 6$.

Example 2: Find the median of the data: 12, 18, 13, 21, 15.

Step-by-step Solution:

First arrange in ascending order: 12, 13, 15, 18, 21. $n = 5$ (odd). Median = term at $(5+1)/2 = 3$ rd term = 15.

WORKED: PIE CHART PERCENTAGE PARTITION



HOTS (Higher Order Thinking Skills) Challenge

Challenge yourself with these complex, multi-concept problems designed to test deeper conceptual understanding and mathematical reasoning:

HOTS Challenge 1: The mean of 5 numbers is 20. If one number is excluded, the mean becomes 18. Find the excluded number.

Analytical Solution Strategy:

Sum of 5 numbers = $5 * 20 = 100$. Sum of 4 numbers = $4 * 18 = 72$. Excluded number = Sum of 5 - Sum of 4 = $100 - 72 = 28$.

HOTS: CENTRAL TENDENCY MEAN VS MEDIAN CURVES



Practice Worksheet

Test your skills by attempting these sample exam problems independently. Draw diagrams where appropriate:

Q1 (1-Mark): Calculate the mean of first five prime numbers.

Q2 (2-Mark): Find the mode of the data: 2, 3, 5, 3, 7, 3, 9, 5.

Q3 (2-Mark): Distinguish between primary and secondary data.

Q4 (5-Mark): Find the median of the numbers: 5, 8, 12, 16, 22, 28 (n is even).

Q5 (5-Mark): Construct a frequency distribution table for the marks of 10 students: 5, 6, 5, 7, 8, 5, 6, 7, 8, 9.

PRACTICE: DATA TALLY FREQUENCY SHEET

Class 9	
Class 10	

Real-World Applications & Connections

This chapter connects directly with modern industrial engineering, computation, and natural systems in numerous ways:

Application Case: Census Statistics

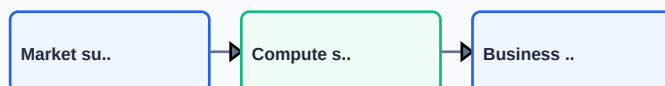
Governments use statistical means and demographics data to plan local education budgets and public infrastructure.

Application Case: Manufacturing Quality Control

Factories analyze production statistics distributions to monitor assembly lines and avoid product defects.

Cross-Curricular Link: Concepts learned in this unit are heavily applied in other subjects like Physics (equations of motion, field vectors) and Data Analytics.

APPLICATION: STATISTICAL ANALYSIS IN MARKETING



Revision Summary & Strategy

Use this checklist to self-evaluate your exam preparation status before final tests:

Key Recall Tips:

- ✓ Always write down 'Given Parameters' first to ensure partial marks are locked in.
- ✓ Pay attention to signs (+ / -) when performing algebraic operations or force mappings.
- ✓ Check that you have written the correct units (e.g. Joules, Tesla, cm^3 , N) for the final numeric answer.
- ✓ Draw neat diagrams using a sharp pencil and ruler for geometry or circuit layouts.

Syllabus Checklist:

- ☐ Read and memorized core definitions on Page 2.
- ☐ Reviewed critical formulas and relations on Page 3.
- ☐ Solved worked examples on Page 4.
- ☐ Attempted worksheet problems on Page 6.

CONGRATULATIONS! YOU HAVE COMPLETED THIS SYLLABUS STUDY GUIDE!

Practice consistently, verify answers with standard textbook methods, and take the topic test in the Nexora Learning app to gauge progress.
Good luck!

REVISION: STATISTICS ANALYTICAL FLOW

